

Video Game Programming: Ethical Aspects

L24

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Game benefits and risks

- Possible benefits:

- Knowledge and skill gain
- Attitude change
- Self-efficacy increase
- Motivation support
- Social inclusion
- Catharsis
- ...

- Possible risks:

- Aggressive behaviors
- Lower school performance
- Stereotyped perceptions
- Media abuse
- Addiction
- ADHD increase
- ...

Violence in games

- Criticism about games typically increases
 - after the release of violent games (e.g., Death Race, Mortal Kombat, Grand Theft Auto, South Park, Rape Day)
 - after youngsters are involved in shootings
- Supporters of violent games
 - say that they are just play and allow for competition and mastery
 - claim freedom of expression
- Opponents of violent games
 - claim that they increase violence in the real world
 - say that they downplay the severity of proposed actions

Implications for developers

- Ethical aspects have **concrete** implications for developers
- **Developer program policies** of game stores such as Google Play and Steam explicitly require to **avoid** some unethical contents (e.g., child endangerment, sexual content, profanity, hate speech, gratuitous violence, dangerous activities and products, sensitive events, bullying, harassment, tobacco, alcohol, illegal drugs)
- Other questioned contents (e.g., fictional violence and nudity for specific purposes) are allowed, but the games should report appropriate **ratings** according to the rules of publishing countries or stores (e.g., PEGI, ESRB, USK, IARC)

Two research perspectives

- A clear distinction between game ethics research studies is difficult
- Egenfeldt-Nielsen, Smith, and Tosca (Understanding Video Games, 2024) present two main perspectives in game ethics research
- **Active Media** perspective believes that media actively influence a mostly passive recipient
- **Active User** perspective stresses the active interpretation and filtering players exhibit when playing video games

Active Media typical research design

- Two groups of subjects: one plays a violent game, the other plays a nonviolent video game
- Before subjects start playing, their level of aggression is measured (e.g., using questionnaires, physiological measures)
- Measurement is repeated after playing
- Any difference in the change in aggression levels between the two groups is attributed to the difference between the two games
- Results are supposed to be generalizable to other violent games played by similar groups

Active User typical research design

- Researchers observe and interview children playing games in natural settings over a period
- They note behaviors, e.g., verbal and physical violence
- They describe the full experience and the context to understand how players integrate video games into daily life
- Effect of playing certain games is just one potential focus
- When incidences of aggression are found, they are analyzed in the specific context in which they arose
- Researchers hesitate to generalize findings because each player perceives each game differently and in different contexts

Active Media foundations

- Based on behaviorist tradition (humans are systems responding to stimuli around them), so media have effects on people
- In-the-lab experimental studies to filter confounding variables, at a given point of time or over a time span
- Considered theories include catharsis (being violent in media reduces internal tensions), cultivation (media distort reality perception), social learning (people imitate models with attached rewards), general arousal (playing increases intensity of actions)
- Models include cognitive neo-association of aggression (violent media lead to aggression by reinforcement of related brain nodes) and general aggression (violent media increase aggressive behaviors by influencing persons' internal state)

Active User foundations

- Based on cultural studies, anthropology, and pedagogy
- Preference for qualitative methods (interviews, field observations) because of personal background and context importance
- Theories include media response (role of person in media interpretation), play as meta communication (play develops ability to communicate that actions depend on context), children's perspective (children and adults have different reference frames with children considering games as a form of play)
- Catalyst model shifts focus on individual disposition for being influenced by violent games with different contributing variables

Active Media and Active User comparison

	<i>Active Media perspective</i>	<i>Active User perspective</i>
Scientific and theoretical starting point	Behaviorism, social psychology, experimental psychology	Anthropology, ethnography, literary studies/semiotics, cultural studies, media theory
Research methods	Predominantly quantitative (controlled studies)	Predominantly qualitative (interview and observation)
Main interest	The effects of media on attitude, perception and behavior	Meaning, role, and function of media
Research object	The video game (what effect the game has on the player)	The player (what the player does with the game)

Active Media studies

- Anderson and Dill, 2000
 - Methods: 220 psychology students, questionnaires about five favorite games, amount of time, aggressiveness / criminal record, academic results
 - Results: males played more violent video games, and spent more time than females; playing violent games correlate with engagement in criminal activities; high playing time hampered academic achievements
- Durkin and Barber, 2002
 - Methods: 1304 adolescents, several variables (family closeness, activity involvement, school engagement, mental health, substance use, self-concept, friendship network, and disobedience)
 - Results: overall, adolescents with low game usage better than those with high and no game usage

Active Media studies

- Robinson et al., 2001
 - Methods: 2 group of children with normal media use and following a course about reducing media use
 - Results: trained group perceived as less aggressive by their friends and showed less verbal aggression
- Gentile et al., 2004
 - Methods: adolescents playing games with more or less violent content; additional variables included gender and involvement of parents
 - Results: adolescents who played more violent games were more hostile, got into more arguments with teachers, and were more involved in physical fights; when researchers included additional variables, the effect of violent game exposures on physical fights practically disappeared

Longitudinal studies

- Longitudinal studies compared aggressiveness after months/years
 - Williams and Skoric, 2005, 213 subjects, no increase after a month
 - Anderson et al., 2008, 1595 subjects, those who played a lot of violent games became more physically aggressive (greater effect on younger)
 - Krahe and Möller, 2010, 1237 subjects, media violence correlated with physical aggression and lower empathy after a year, but small effect size
 - Ferguson et al., 2011, 165 subjects, no relation or negative outcomes based on exposure to violent video games after 1 and 3 years
 - Willoughby et al., 2012, 1492 subjects, violent games predicted higher aggression, whereas nonviolent games did not
 - Kühn et al., 2018, 90 subjects, no link found
 - Przybylski and Weinstein, 2019, 1000 subjects, no link found

Meta-studies

- Very different conclusions
 - Griffiths, 1999: only consistent finding is that young children are affected by violence more than teenagers and older children
 - Anderson and Bushman, 2001: violent video games increase aggressive behavior in children and young adults
 - Sherry, 2001: effect of games smaller than that of violent television
 - Freedman, 2001: too little evidence to reach any firm conclusion
 - Prescott et al., 2018: strongest link with white participants, less strong among Asian ones, and not reliably for Hispanic ones; small effect size explaining only around 1 percent of variance

Criticism of Active Media

- Laboratory vs everyday situations
- Problems of causality
- Defining aggression
- Games are a rich and varied phenomenon
- Statistical problems
- Criticism of basic assumptions

Active User studies

- Sørensen and Jessen, 2000:
 - Methods: 31 children observed and interviewed to understand ability to distinguish between fiction and reality
 - Results: negative influence from violent games smaller than other media
- Livingstone, 2002:
 - Methods: 15000 children, qualitative and quantitative methods
 - Results: media plays a central role in homes and games are prominent

Active User studies

- Turkle, 1995: technology can have an effect but that this is probably neither positive nor negative
- Ermi and Mäyrä, 2003: games can captivate children through features such as challenges, vast virtual worlds, and freedom
- Olson, 2010: compared with other media, games appear to have an unusually expansive appeal and serve a surprising number of emotional, social, and intellectual needs

Criticism of Active User

- There is a limited number of Active User studies
- Active User studies rarely apply rigorous methods
- Observations of a group of children's reactions toward violence do not necessarily apply to other children

Game addiction

- In the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), video game addiction was included under the classification Internet Gaming Disorder
- Addiction to video games does not have the same repercussions as other forms of addiction
- Griffiths: percentage of addicted players ranges from 7%, to 19.9% to 37.5% in different studies
- Gentile: 8.5%-9% of players exhibited pathological patterns
- Meta-studies by Kush and Griffiths, 2012 and Strittmatter et al., 2015 confirm that there is addiction in 3.1% to 3.62% of players

Conclusions

- Over the last 35 years, more than 200 studies have been conducted to understand relationship between games and risk
- Both presented perspectives have strengths and weaknesses
- Attempts to combine perspectives are being made
- Positive and negative effects of games have been shown
- Effects vary with game content and players
- **Game programmers have a responsibility about the content they put in their games**

Bibliografy

- Egenfeldt-Nielsen, Smith, Tosca, Understanding Video Games: The Essential Introduction, 5th ed., Routledge, 2024

